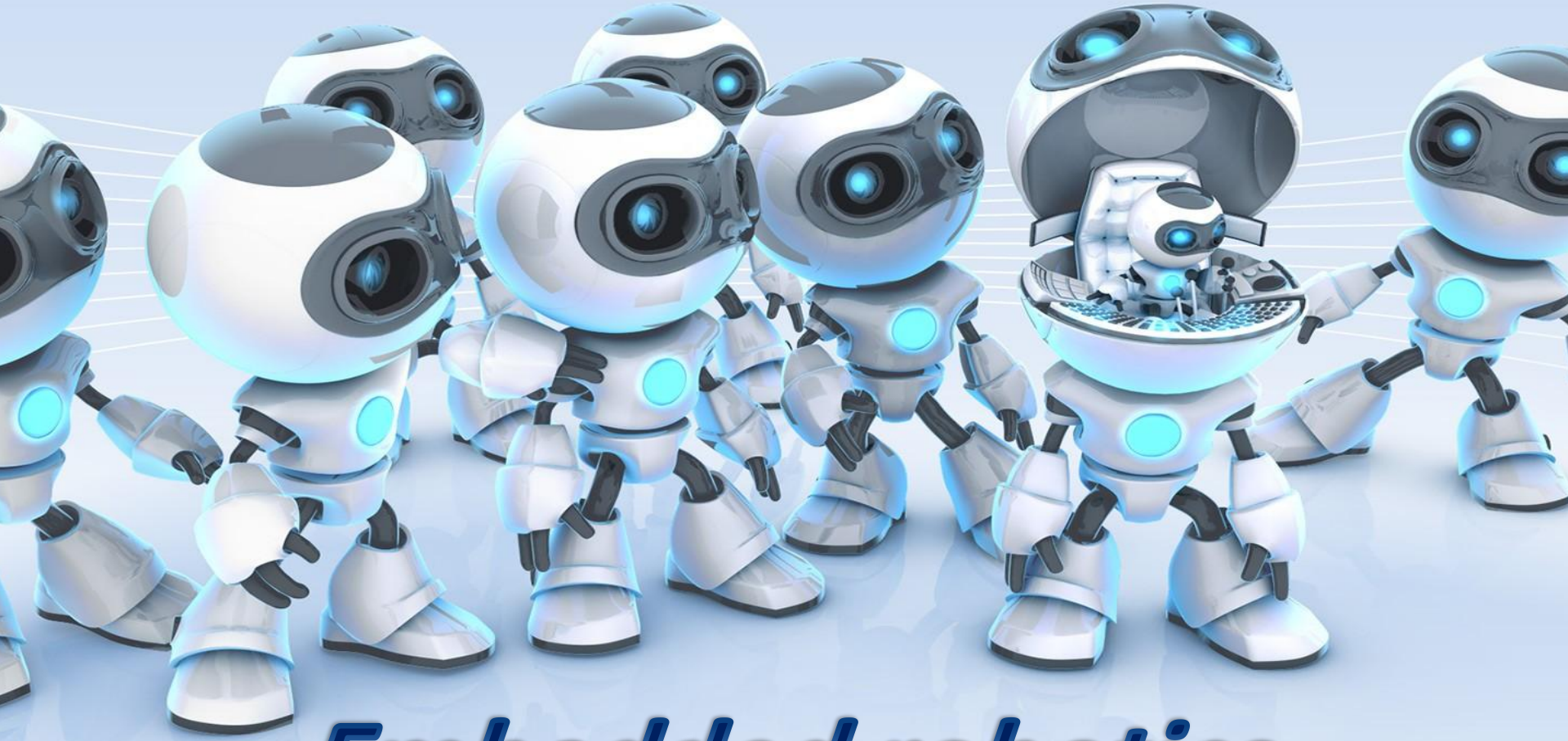


Embedded robotics

In jedem steckt ein Büronau

Introduction



- Robotics has come a long way.
- Especially for mobile robots, a similar trend is happening as we have seen for computer systems, the transition from mainframe computing via workstations to PCs, which will probably continue with handheld devices

ESET Smart Security

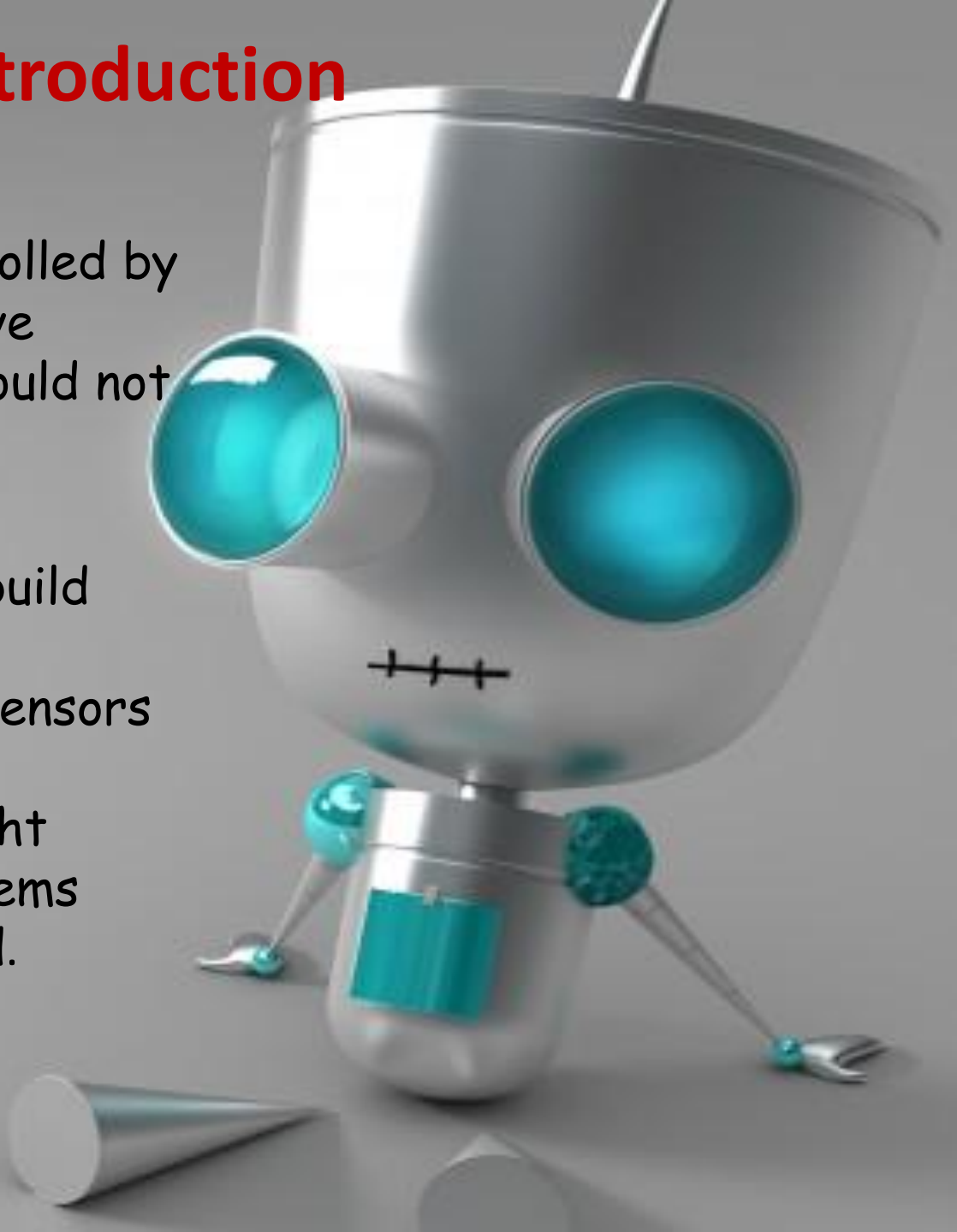
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Introduction

In the past

Mobile robots were controlled by heavy, large, and expensive computer systems that could not be carried

Today, however, we can build small mobile robots with numerous actuators and sensors that are controlled by inexpensive, small, and light embedded computer systems that are carried on-board.





Introduction

Mobile robot used or found in many fields

It is considered as a perfect tool for engineering, education…

Mobile robots are use today at almost all universities in :

1) Computer Science
2) Computer Engineering
3) Information Technology
4) Electrical Engineering
5) Mechanical Engineering

What are the advantages of using mobile robot systems as opposed to traditional ways of education, for example mathematical models or computer simulation?

- First of all, a robot is a tangible, self-contained piece of real-world hardware.
- Students can relate to a robot much better than to a piece of software.
- Tasks to be solved involving a robot are of a practical nature and directly "make sense" to students, much more so than, for example, the inevitable comparison of sorting algorithms.



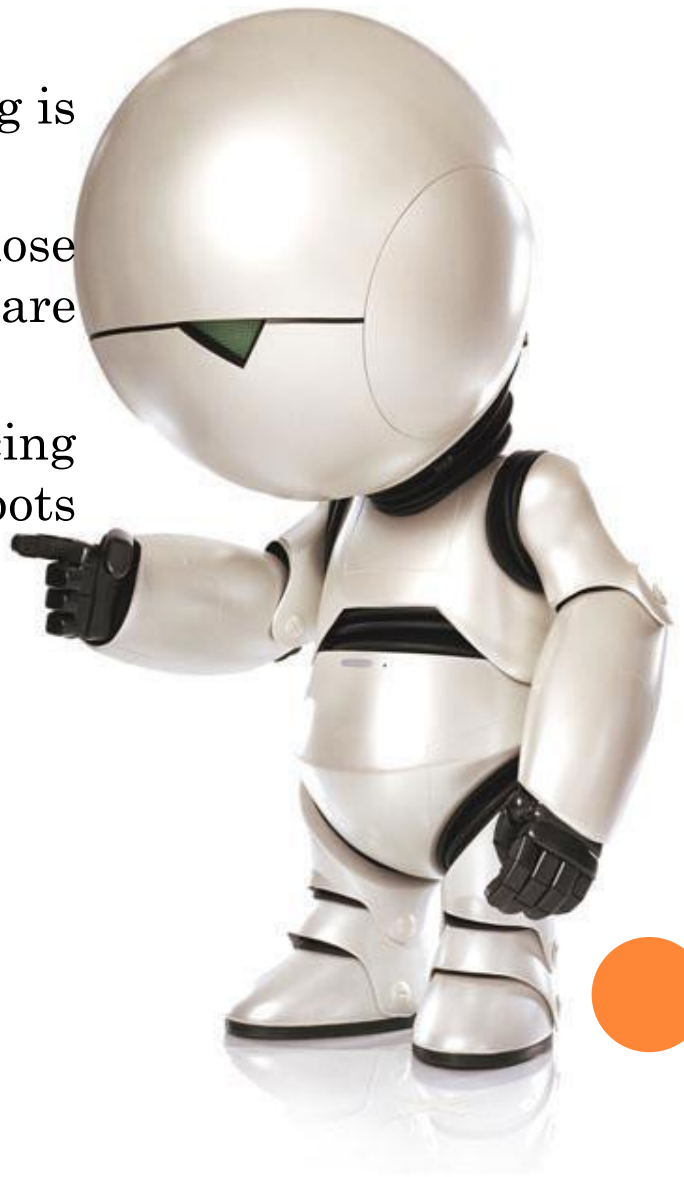
What are the advantages of using mobile robot systems as opposed to traditional ways of education, for example mathematical models or computer simulation?

- Secondly, all problems involving "real-world hardware" such as a robot, are in many ways harder than solving a theoretical problem.
- The "perfect world" which often is the realm of pure software systems does not exist here.
- Any actuator can only be positioned to a certain degree of accuracy, and all sensors
- have intrinsic reading errors and certain limitations.
- Therefore, a working robot program will be much more than just a logic solution coded in software.
- It will be a robust system that takes into account and overcomes inaccuracies and imperfections.



What are the advantages of using mobile robot systems as opposed to traditional ways of education, for example mathematical models or computer simulation?

- Third and finally Mobile Robot Programming is enjoyable and an inspiration to students
- The fact that there is a moving system whose behavior can be specified by a piece of software is a challenge.
- This can even be amplified by introducing robot competitions where two teams of robots compete in solving a particular task



Mobile Robots



We have developed a num

1) Wheeled Robot
2) Tracked Robot
3) Legged Robot
4) Flying Robot
5) Underwater Robot

We call these robots the "Eye
robots because they are all us
controller "EyeCon"





Wheeled Robot



Tracked Robot



Legged Robot



Underwater Robot



Flying Robot

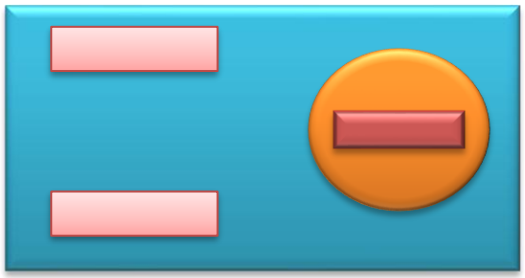
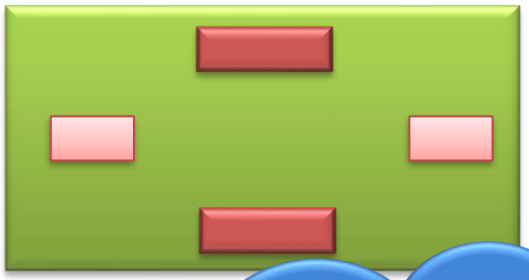
Wheeled Robots

- The simplest case of mobile robots are wheeled robots.
- Wheeled robots comprise one or more *driven wheels* and have optional *passive* or *caster wheels* and possibly *steered wheels*.
- Most designs require *two motors* for driving (and steering) a mobile robot.



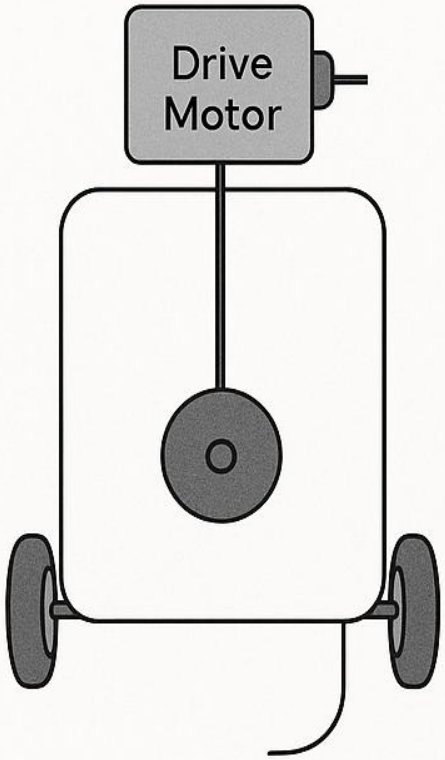


Wheeled Robots

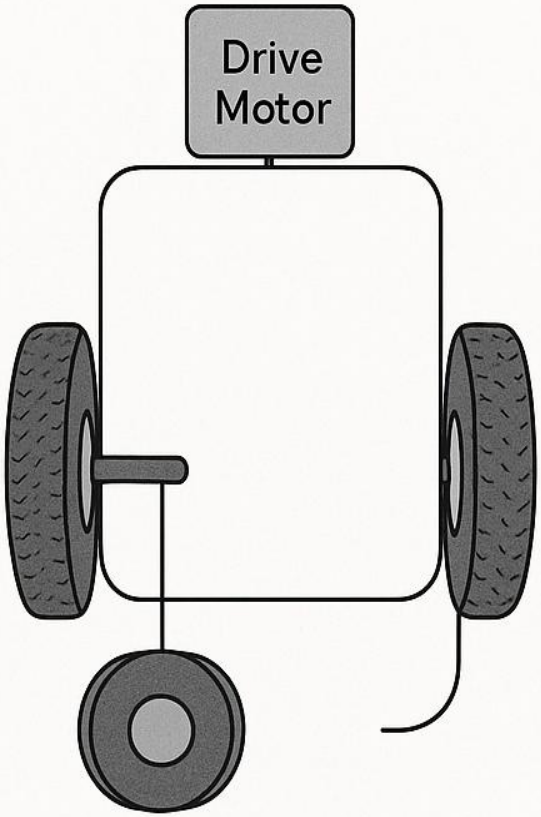




- The design on the left-hand side … also steered.
- It requires two motors, one for …
- The advantage of this design i… …ve been *completely separated* by us…
- A disadvantage of this des… …ince the driven wheel is not located a…
- The robot design in the mid… …ne of the most commonly used mobile robot…
- The combination of two driven wh… …o be driven straight, in a curve, or to turn on the spot.
- The translation between driving comman…s and the corresponding wheel speeds has to be done using software. (video)
- Finally, on the right-hand side is the so-called "*Ackermann Steering*", which is the standard drive and steering system of a rear-driven passenger car.
- We have one motor for driving both rear wheels via a differential box and one motor for combined steering of both front wheels (video).

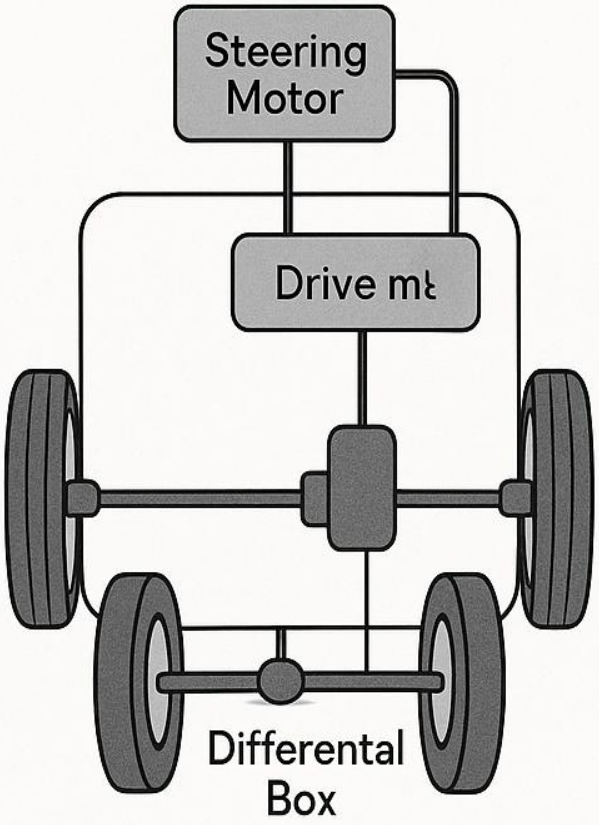
It is interesting to note that all of these different mobile robot designs require two motors in total for driving and steering.



Single Driven Wheel

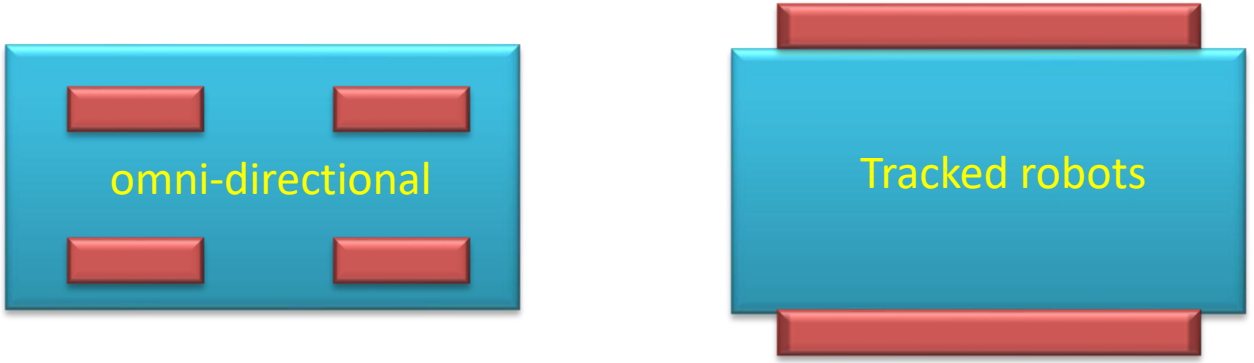


Differential Drive

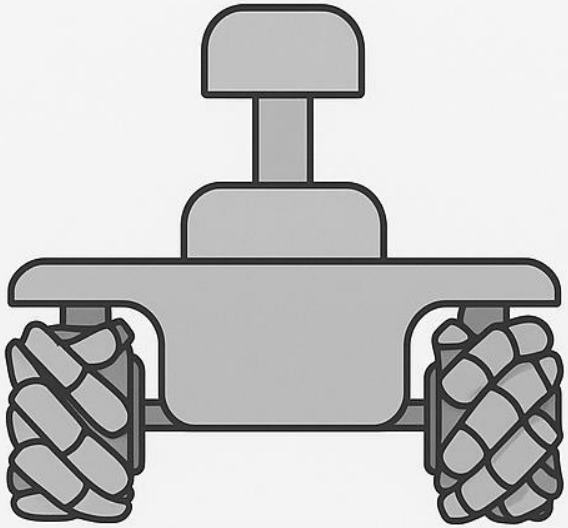


Ackermann Steering

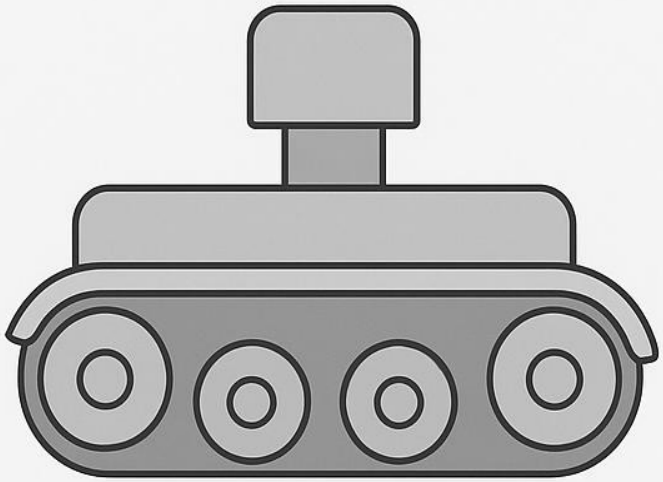
Wheeled Robots



- A special case of a wheeled robot is the *omni-directional* "Mecanum drive" robot in Figure, left.
- It uses four driven wheels with a special wheel design.
- One disadvantage of all wheeled robots is that they require a street or some sort of flat surface for driving. (video)
- Tracked robots (middle) are more flexible and can navigate over rough terrain.
- However, they cannot navigate as accurately as a wheeled robot.
- Tracked robots also need two motors, one for each track. (video)



Mecanum drive



Tracked

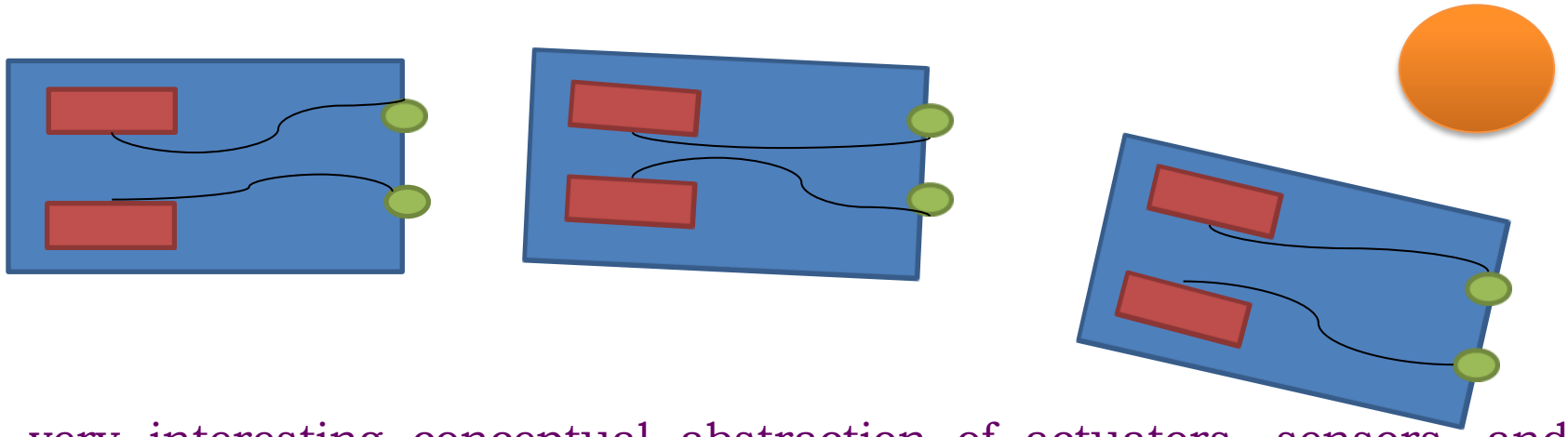
Legged Robots



- Legged robots are the final category of land-based mobile robots.
- Like tracked robots, they can navigate over rough terrain or climb up stairs.
- There are many designs for legged robots, depending on the *number of legs*.
- The general rule is the more legs, the easier to *balance*.
- For example, the six-legged robot shown can be operated in such a way that three legs are always on the ground while three legs are in the air.
- The robot will be stable at all times, *resting on a tripod* formed from the three legs currently on the ground (provided its center of mass falls in the triangle described by these three leg
- Legged robots usually require two or more motors ("degrees of freedom") *per leg*, so a six legged robot requires at least 12 motors.
- A biped (two-legged) robot cannot play the same trick with a supporting triangle, So other techniques for balancing need to be employed, as is discussed later
- Many biped robot designs have *five or more motors per leg*, which results in a rather large total number of degrees of freedom and also in considerable *weight and cost*.



Sensor-Actuator-Controller Interaction

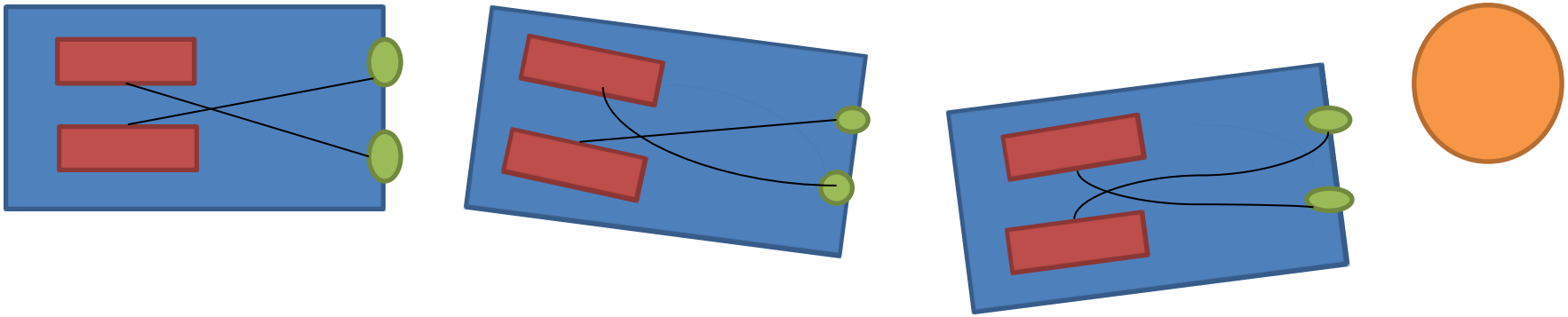


A very interesting conceptual abstraction of actuators, sensors, and robot control is the vehicles described by Braitenberg [Braitenberg 1984].

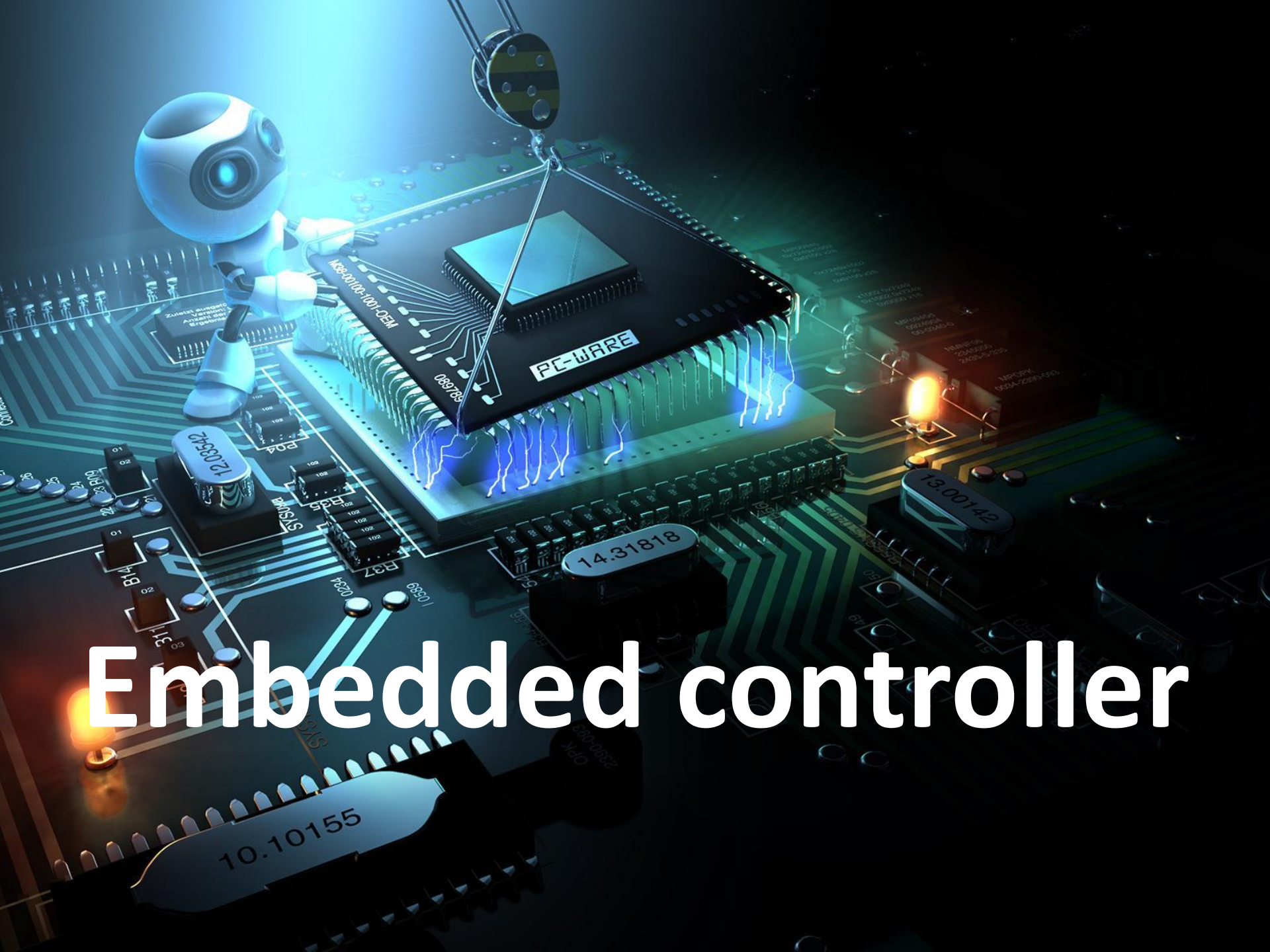
For example, we have a simple interaction between motors and light sensors.

- If a light sensor is activated by a light source, it will proportionally increase the speed of the motor it is linked to.
- Our robot has two light sensors, one on the front left, one on the front right.
- The left sensor is linked to the left motor, the right sensor to the right motor.
- If a light source appears in front of the robot, it will start driving toward it, because both sensors will activate both motors.
- However, what happens if the robot gets closer to the light source and goes slightly off course?
- In this case, one of the sensors will be closer to the light source, and therefore one of the motors will become faster than the other.
- This will result in a curve trajectory and it will miss the light source.

Sensor-Actuator-Controller Interaction



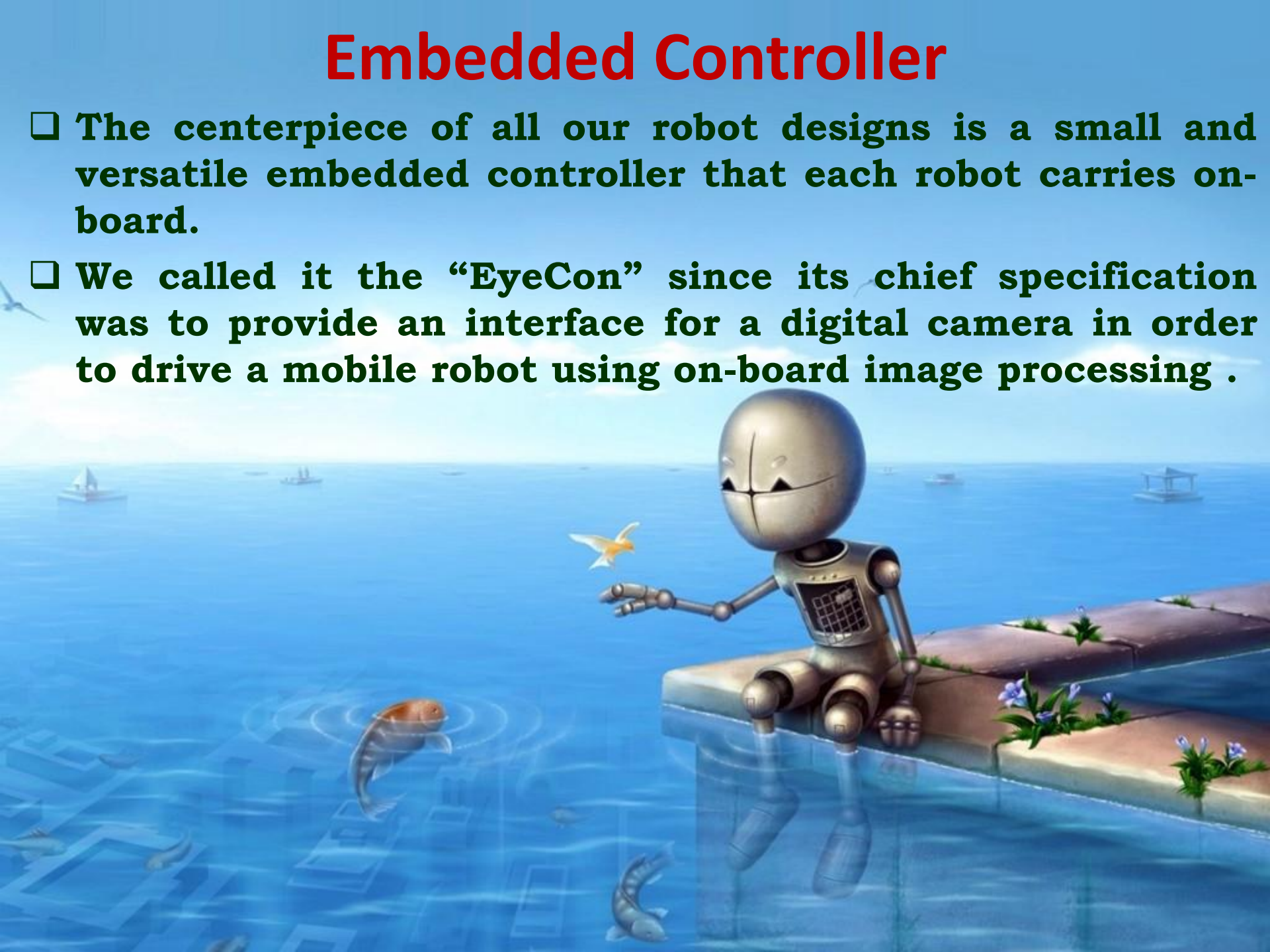
- ✓ Figure shows a very similar scenario of Braitenberg vehicles. However, here we have linked the left sensor to the right motor and the right sensor to the left motor.
- ✓ If we conduct the same experiment as before, again the robot will start driving when encountering a light source.
- ✓ But when it gets closer and also slightly off course (veering to the right in the figure), the left sensor will now receive more light and therefore accelerate the right motor.
- ✓ This will result in a left curve, so the robot is brought back on track to find the light source.



Embedded controller

Embedded Controller

- The centerpiece of all our robot designs is a small and versatile embedded controller that each robot carries on-board.
- We called it the “EyeCon” since its chief specification was to provide an interface for a digital camera in order to drive a mobile robot using on-board image processing .



The EyeCon is a small, light, and fully self-contained embedded
The main characteristics of the EyeCon are:

- 25MHz 32bit controller (Motorola M68332)
- 1MB RAM, extendable to 2MB
- 512KB ROM (for system + user programs)
- 1 Parallel port
- 3 Serial ports (1 at V24, 2 at TTL)
- 8 Digital inputs
- 8 Digital outputs
- 16 Timing processor unit inputs/outputs
- 8 Analog inputs
- Single compact PCB
- Interface for color and grayscale camera
- Large graphics LCD (128× 64 pixels)controller
- Reset button
- Power switch
- Audio output
- Piezo speaker
- Adapter and volume potentiometer for external speaker
- Microphone for audio input
- Battery level indication
- Connectors for actuators and sensors:
- Digital camera
- 2 DC motors with encoders
- 12 Servos
- 6 Infrared sensors
- 6 Free analog inputs

Interfaces -

- A number of interfaces are available on most embedded systems.
- These are digital inputs, digital outputs, and analog inputs.
- Analog outputs are not always required and would also need additional amplifiers to drive any actuators.

